Today

- Commutation relations
- Time-independent Schrödinger equation

1.13 Commutation relations between position and momentum

We have already discussed that the position and the momentum operators are given by

$$\hat{\vec{r}} = \vec{r}, \qquad \hat{\vec{p}} = -i\hbar\vec{\nabla}. \tag{1.13.1}$$

Now consider applying $\hat{x}$ first and $\hat{p}_x$ second on the wavefunction $\psi(x,t)$,

$$\hat{p}_x\hat{x}\psi(x,t) = \hat{p}_x\left(\hat{x}\psi(x,t)\right) = -i\hbar\frac{\partial}{\partial x}\left(x\psi(x,t)\right) = -i\hbar\psi(x,t) - i\hbar x\frac{\partial}{\partial x}\psi(x,t). \tag{1.13.2}$$

Next, suppose we reverse the order of operations by applying $\hat{p}_x$ first and $\hat{x}$ second,

$$\hat{x}\hat{p}_x\psi(x,t) = \hat{x}\left(\hat{p}_x\psi(x,t)\right) = x\left(-i\hbar\frac{\partial}{\partial x}\psi(x,t)\right) = -i\hbar x\frac{\partial}{\partial x}\psi(x,t). \tag{1.13.3}$$

Thus, we see that $\hat{p}_x\hat{x}\psi(x,t) \neq \hat{x}\hat{p}_x\psi(x,t)$, and

$$\left(\hat{x}\hat{p}_x - \hat{p}_x\hat{x}\right)\psi(x,t) = i\hbar\,\psi(x,t) \tag{1.13.4}$$

Denoting $\hat{x}\hat{p}_x - \hat{p}_x\hat{x}$ by $[\hat{x},\,\hat{p}_x]$, called the commutator between $\hat{x}$ and $\hat{p}_x$,

$$[\hat{x},\,\hat{p}_x] = i\hbar. \tag{1.13.5}$$

Because $[\hat{x},\,\hat{p}_x] \neq 0$, we say $\hat{x}$ and $\hat{p}_x$ do not commute.

On the other hand,

$$\begin{aligned}[\hat{x},\,\hat{p}_y]\psi(\vec{r},t) &= x(-i\hbar\frac{\partial}{\partial y})\psi(\vec{r},t) - (-i\hbar\frac{\partial}{\partial y}(x\psi(\vec{r},t))) \\ &= x(-i\hbar\frac{\partial}{\partial y})\psi(\vec{r},t) + x(i\hbar\frac{\partial}{\partial y}(\psi(\vec{r},t))) = 0,\end{aligned} \tag{1.13.6}$$

and $\hat{x}$ and $\hat{p}_y$ commute.

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Similarly and collecting,

$$[\hat{x},\, \hat{p}_x] = [\hat{y},\, \hat{p}_y] = [\hat{z},\, \hat{p}_z] = i\hbar \tag{1.13.7}$$

and the components of $\hat{\vec{r}}$ and $\hat{\vec{p}}$ commute for any other combination, such as

$$[\hat{x},\, \hat{p}_y] = [\hat{x},\, \hat{y}] = [\hat{p}_x,\, \hat{p}_y] = 0. \tag{1.13.8}$$

We will investigate later that when two operators do not commute there is uncertainty in measuring them simultaneously.

Denoting the components of the vectors by numerical indices with $x = x_1$, $y = x_2$, $z = x_3$, $p_x = p_1$, $p_y = p_2$, and $p_z = p_3$, the commutation relations among the components of $\hat{\vec{r}}$ and $\hat{\vec{p}}$ can be summarized as

$$[\hat{x}_i,\, \hat{p}_j] = i\hbar\delta_{ij}, \quad [\hat{x}_i,\, \hat{x}_j] = [\hat{p}_i,\, \hat{p}_j] = 0, \tag{1.13.9}$$

where δ_{ij} is the Kronecker delta defined as

$$\delta_{ij} = \begin{cases} 1, & i = j \\ 0, & i \neq j. \end{cases}$$

Exercise 1.13.1: Determine $[\hat{x},\, \hat{p}_x^2]$.

Solution: $[\hat{x},\, \hat{p}_x^2] = \hat{x}\hat{p}_x^2 - \hat{p}_x^2\hat{x} = \hat{x}\hat{p}_x\hat{p}_x - \hat{p}_x\hat{p}_x\hat{x} = \hat{x}\hat{p}_x\hat{p}_x - \hat{p}_x\hat{x}\hat{p}_x + \hat{p}_x\hat{x}\hat{p}_x - \hat{p}_x\hat{p}_x\hat{x} = [\hat{x},\, \hat{p}_x]\hat{p}_x + \hat{p}_x[\hat{x},\, \hat{p}_x] = i\hbar\hat{p}_x + \hat{p}_x i\hbar = 2i\hbar\hat{p}_x$.

In general, given three operators $\hat{A}$, $\hat{B}$, and $\hat{C}$, we have $[\hat{A},\, \hat{B}\hat{C}] = \hat{A}\hat{B}\hat{C} - \hat{B}\hat{C}\hat{A} = \hat{A}\hat{B}\hat{C} - \hat{B}\hat{A}\hat{C} + \hat{B}\hat{A}\hat{C} - \hat{B}\hat{C}\hat{A}$, where we have subtracted and added $\hat{B}\hat{A}\hat{C}$. Therefore, $[\hat{A},\, \hat{B}\hat{C}] = [\hat{A},\, \hat{B}]\hat{C} + \hat{B}[\hat{A},\, \hat{C}]$. Thus,

$$[\hat{A},\, \hat{B}\hat{C}] = \hat{B}[\hat{A},\, \hat{C}] + [\hat{A},\, \hat{B}]\hat{C}. \tag{1.13.10}$$

1.14 Time-independent Schrödinger equation

Suppose the potential energy is independent of time; i.e., $V(\vec{r}, t) = V(\vec{r})$. The Schrödinger equation then becomes

$$i\hbar\frac{\partial\psi(\vec{r}, t)}{\partial t} = -\frac{\hbar^2}{2m}\vec{\nabla}^2\psi(\vec{r}, t) + V(\vec{r})\psi(\vec{r}, t), \tag{1.14.1}$$

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Because the potential does not mix position and time, we can use the method of separation of variables and start with the ansatz

$$\psi(\vec{r},t) = \psi(\vec{r})\phi(t). \tag{1.14.2}$$

With this, (1.14.1) becomes

$$i\hbar\psi(\vec{r})\frac{d\phi(t)}{dt} = -\phi(t)\frac{\hbar^2}{2m}\vec{\nabla}^2\psi(\vec{r}) + V(\vec{r})\phi(t)\psi(\vec{r}). \tag{1.14.3}$$

Dividing both sides by $\psi(\vec{r})\phi(t)$,

$$i\hbar\frac{1}{\phi(t)}\frac{d\phi(t)}{dt} = \frac{1}{\psi(\vec{r})}\left[-\frac{\hbar^2}{2m}\vec{\nabla}^2\psi(\vec{r}) + V(\vec{r})\right]. \tag{1.14.4}$$

Because the left hand side is a function of only t and the right hand side is a function of only $\vec{r}$, each side must be a constant.

Question 1.14.1: Verify why when functions of two independent variables are equal, they must be a constant.

Answer: Consider two functions, $f(x)$ and $g(y)$, where x and y are independent variables. If $f(x) = g(x)$, then taking a derivative of both sides with respect to x, $\frac{df(x)}{dx} = \frac{dg(y)}{dx} = 0$. Therefore, $\frac{df(x)}{dx} = 0$, and $f(x)$ is a constant independent of x. Similarly, $g(y)$ is a constant. Thus, $f(x) = g(x) = const.$

Let the constant be denoted by E. The equation of motion for $\phi(t)$ is then

$$i\hbar\frac{d\phi(t)}{dt} = E\phi(t), \tag{1.14.5}$$

and the solution is

$$\phi(t) = e^{-\frac{i}{\hbar}Et}, \tag{1.14.6}$$

where we have set the constant coefficient to one, since any constant can be absorbed in $\psi(\vec{r})$ and the total wavefunction needs to be normalized at the end.

The equation of motion of $\psi(\vec{r})$ is

$$-\frac{\hbar^2}{2m}\vec{\nabla}^2\psi(\vec{r}) + V(\vec{r})\psi(\vec{r}) = E\psi(\vec{r}), \tag{1.14.7}$$

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or

$$\hat{H}\psi(\vec{r}) = E\psi(\vec{r}), \tag{1.14.8}$$

where $\hat{H} = -\frac{\hbar^2}{2m}\vec{\nabla}^2 + V(\vec{r})$ is the Hamiltonian operator. This is called the time-independent Schrödinger equation.

Question 1.14.2: Does the time-independent Schrödinger equation imply that the wavefunction of a system with time-independent potential is time-independent?

Answer: No. It means that $\psi_E(\vec{r},t) = \psi_E(\vec{r})e^{-\frac{i}{\hbar}Et}$, where $\psi_E(\vec{r})$ is the solution to the time-independent Schrödinger equation with the constant E. In other words, the time dependence for a state of constant E is given by the constant phase factor of $e^{-\frac{i}{\hbar}Et}$.

Observe that when the potential is time-independent, the time dependence of the wavefunction that corresponds to the constant E is easily written once we have solved the time-independent Schrödinger equation,

$$\psi_E(\vec{r},t) = \psi_E(\vec{r})e^{-\frac{i}{\hbar}Et}, \tag{1.14.9}$$

where we have added subscript E to identify $\psi_E(\vec{r},t)$ as a wavefunction that corresponds to constant E.

Exercise 1.14.1: The wavefunction of a particle of mass m in a one-dimensional space is given to be $\psi(x,t) = Ae^{-\frac{x^2}{4\sigma^2}-\frac{i}{\hbar}Et}$, where A, σ, and E are real constants.

1. Write down the corresponding normalized wavefunction.
2. Determine the probability for finding the particle between $x = 0$ and $x = \infty$. (5 points)
3. Determine $\langle\hat{x}\rangle$, $\langle\hat{x}^2\rangle$, and $\Delta x \equiv \sqrt{\langle\hat{x}^2\rangle - \langle\hat{x}\rangle^2}$.
4. Determine the potential energy of the particle in terms of given parameters. Show that you obtain the standard simple harmonic potential $V(x) = \frac{1}{2}m\omega^2x^2$ for $E = \frac{1}{2}\hbar\omega$ and $\sigma = \sqrt{\frac{\hbar}{2m\omega}}$.

1.15 Properties of solutions for time-independent potentials

Since we will be investigating only time-independent potentials in nearly all of this course, it is important to summarize some important properties of the solutions when the potential is time-independent.

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1.15.1 The constant E must be real

One way that the reality of E follows is from Born's hypothesis on the normalizability of the wavefunction. In particular, considering $\psi_E(\vec{r},t) = \psi_E(\vec{r})\, e^{-\frac{i}{\hbar}Et}$ and writing $E = E_R + iE_I$, where E_R and E_I are possible real and imaginary parts of E, we have $|\psi_E(\vec{r},t)|^2 = |\psi_E(\vec{r})|^2 e^{\frac{i}{\hbar}(E_R - iE_I)t}$ $e^{-\frac{i}{\hbar}(E_R+iE_I)t} = |\psi_E(\vec{r})|^2 e^{\frac{2}{\hbar}E_I t}$, and

$$\int |\psi_E(\vec{r},t)|^2 d^3\vec{r} = \int |\psi_E(\vec{r})|^2 d^3\vec{r} e^{\frac{2}{\hbar}E_I t}, \tag{1.15.1}$$

which cannot be set equal to one for all t except when $E_I = 0$. Therefore, E must be real.

1.15.2 Stationary wavefunction

Given the wavefunction $\psi_E(\vec{r},t)$ that corresponds to E,

$$\mathcal{P}(\vec{r},t) = |\psi_E(\vec{r},t)|^2 = |\psi_E(\vec{r})|^2, \tag{1.15.2}$$

and the probability density is independent of time. We say the wavefunction $\psi_E(\vec{r},t)$ is stationary.

> **Question 1.15.1:** The uncertainty of the energy ΔE in a stationary state of energy E is (A). $\frac{\hbar}{2}$, (B). $\hbar$, (C). 0, (D). E, (E). $2E$.
>
> **Answer:** The answer is (C). 0. That is because stationary state is a state of definite energy. See below for more detail.

1.15.3 Stationary wavefunction is a state of definite energy

Using $\hat{H}\psi_E(\vec{r},t) = E\psi_E(\vec{r},t)$, the expectation value of the Hamiltonian operator in state $\psi_E(\vec{r},t)$ is

$$\langle \hat{H} \rangle = \int \psi_E(\vec{r},t)^* \hat{H} \psi_E(\vec{r},t) d^3\hat{r} = \int \psi_E(\vec{r},t)^* E \psi_E(\vec{r},t) d^3\hat{r} = E \int |\psi_E(\vec{r},t)^2 d^3\hat{r} = E. \tag{1.15.3}$$

Therefore, the constant E is the expectation value of the Hamiltonian operator, and it is energy. Furthermore,

$$\langle \hat{H}^2 \rangle = \int \psi_E(\vec{r},t)^* \hat{H}^2 \psi_E(\vec{r},t) d^3\hat{r} = \int \psi_E(\vec{r},t)^* EE \psi_E(\vec{r},t) d^3\hat{r} = E^2 \int |\psi_E(\vec{r},t)^2 d^3\hat{r} = E^2, \tag{1.15.4}$$

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and the uncertainty or variance of the energy is

$$\Delta E = \sqrt{\langle \hat{H}^2 \rangle - \langle \hat{H} \rangle^2} = \sqrt{E^2 - E^2} = 0. \tag{1.15.5}$$

Therefore, the stationary state $\psi_E(\vec{r}, t)$ is a state of definite energy E.

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